DEMO-GDE-006 Synthetic Loan Portfolio Concentration and Exposure Review Guide

Version: 1.0

Effective date: 1 January 2025

Owner: Synthetic Portfolio Exposure Analytics Team

Classification: Synthetic Demo Only

1. Purpose and Scope

- This guide defines exposure concentration analysis using `loan_quality` only.
- It complements the delinquency monitoring procedure in DEMO-SOP-002.
- Each row represents one fictional loan observation, identified by `loan_id`.
- `assessment_date` sets the date scope for every concentration report.
- The available reporting period is 1 January 2025 through 30 September 2026.
- Regions include Bali, DKI Jakarta, Jawa Barat, Jawa Tengah, Jawa Timur, and Sumatera Utara.
- `customer_segment` includes Retail, SME, and Priority.
- `origination_channel` includes Digital, Branch, and Partner.
- `product_category` includes Personal Loan, Working Capital, and Micro Loan.

2. Governed Exposure Measures

- Total Outstanding equals `SUM(outstanding_balance)` for the selected loan observations.
- Observed Loans equals `COUNT(*)` within the same filter scope.
- Average Outstanding equals `AVG(outstanding_balance)` within the scope.
- Delinquent Balance equals `SUM(CASE WHEN days_past_due > 30 THEN outstanding_balance ELSE 0 END)`.
- Delinquent Balance Share is Delinquent Balance divided by Total Outstanding, multiplied by 100.
- A zero Total Outstanding denominator must result in a null or unavailable rate, not an invented percentage.
- Monthly Installment Total equals `SUM(monthly_installment)` and is descriptive only.
- Currency measures are expressed in IDR.
- No measures in this section depend on `credit_applications` or a table join.

3. Concentration Analysis

- The analyst may group Total Outstanding by `region_name`, `customer_segment`, `product_category`, or `origination_channel`.
- Exposure Share for a group is its Total Outstanding divided by Total Outstanding for the declared comparison population, multiplied by 100.
- A valid exposure-share comparison must use one dataset, one date period, and explicitly stated group membership.
- Do not sum overlapping subgroup shares and claim a portfolio total.
- High outstanding exposure alone does not demonstrate elevated delinquency or credit losses.
- Delinquent Balance Share is not equivalent to the regulatory NPL ratio.
- The table contains no provisions, write-offs, collateral valuations, or regulatory classifications.

4. Exposure and Delinquency Review

- Delinquent means `days_past_due > 30`.

- `risk_band` is Low for 0 days, Medium for 1–30 days, and High for more than 30 days.
- An investigation trigger requires a delinquency rate greater than 15 percent and at least 30 Observed Loans in the same group.
- Both conditions must be satisfied before labeling a group as meeting the synthetic monitoring trigger.
- The trigger should be reported with the numerator and denominator.
- Concentration patterns and delinquency patterns are separate observations.
- Neither pattern proves a customer, channel, or region caused a risk outcome.

5. Data Quality and Communication

- Check that `outstanding_balance` and `monthly_installment` are nonnegative.
- Confirm that `delinquency_flag` matches the rule `days_past_due > 30`.
- Confirm that `risk_band` agrees with the defined day ranges.
- The dataset has one observation per `loan_id` rather than repeated loan-month snapshots.
- The assistant must not describe changes in the same loan through time.
- Analysts must identify small group sizes and avoid confident causal claims.
- Findings should be escalated to a human risk owner for verification and action.

Disclaimer

This document is fictional. It is not an actual bank policy or an OJK regulation. It must not be used for real credit decisions. No automated credit decision is permitted. The data contain no real customer names, NIK, card numbers, or other PII. The assistant must distinguish observed association from confirmed causation.